

Project Title: Student Result Management System

Objective

Build a simple Python console application using the topics you have learned this week.

Project Requirements

Create a Python program that performs the following tasks:

1. Student Information

- Ask the user to enter:
 - Student Name
 - Student ID
 - Department

Store the information using appropriate variables.

2. Subject Marks

Create a list of the following subjects:

`["Python", "Math", "English", "Physics", "ICT"]`

Using a **for loop**, ask the user to enter marks for each subject.

Store all marks in a dictionary like:

```
{  
    "Python": 90,  
    "Math": 85,  
    ...  
}
```

3. Calculate Result

Calculate

- Total Marks
- Average Marks
- Highest Mark
- Lowest Mark

Use Python's built-in math functions where possible.

4. Grade Calculation

Using **if-elif-else**, assign grades.

Average	Grade
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80-100	A+
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70-79	A
-------	---

60-69	A-
-------	----

50-59	B
-------	---

40-49	C
-------	---

Below 40	F
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5. Pass or Fail

If any subject mark is below **40**, display

Status: Failed

Otherwise

Status: Passed

6. Password Verification

Ask the user to enter a password.

Correct password:

python123

Use a **while loop** until the correct password is entered.

7. String Operations

Display:

- Student name in uppercase
 - Student name in lowercase
 - Length of the student name
 - First three characters
 - Last three characters
-

8. Set Example

Create two sets.

```
sports = {"Football", "Cricket", "Badminton"}
```

```
clubs = {"Programming", "Cricket", "Photography"}
```

Display

- Common items
 - All unique items
-

9. Tuple Example

Create a tuple containing weekdays.

Display

- First day
 - Last day
 - Total number of days
-

10. Final Report

Display the report like this:

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STUDENT REPORT

=====

Name : Rahim
ID : 221-15-001
Department: CSE

Python : 90
Math : 85
English : 80
Physics : 95
ICT : 88

Total : 438
Average : 87.6
Highest : 95
Lowest : 80

Grade : A+
Status : Passed

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Bonus Challenge (Optional)

Try to use:

- List Comprehension
 - Nested Loop
 - continue statement
 - break statement
 - Formatted Strings (`f" "`)
-

Submission Rules

- Use meaningful variable names.
- Keep the code properly formatted.

- Do not copy from others.
- Submit a single Python (.py) file.